

Chapter 1

Deterministic and Stochastic Correlation

1.1 Outcomes and Probabilities of Outcomes

1.1.1 Normal Distribution

The **normal distribution** has a probability density function (PDF) of

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma^2} \exp \left[\frac{-(x - \mu)^2}{2\sigma^2} \right]. \quad (1.1)$$

This PDF is valid since it is non-negative for values of support $x \in \mathbb{R}$, and because the integral from $-\infty$ to ∞ is equal to unity. The **cumulative distribution** function (CDF) is

$$P(x) = \int_{-\infty}^x p(\tau) d\tau = \frac{1}{2} \left(1 + \operatorname{erf} \left[\frac{x - \mu}{\sqrt{2}\sigma} \right] \right). \quad (1.2)$$

As $x \rightarrow -\infty$, $P \rightarrow 0$. As $x \rightarrow \infty$, $P \rightarrow 1$. The **mean value** is

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} p(x) \cdot x dx \quad (1.3)$$

$$= \int_{-\infty}^{\infty} \frac{x}{\sqrt{2\pi}\sigma^2} \exp \left[\frac{-(x - \mu)^2}{2\sigma^2} \right] dx \quad (1.4)$$

$$= \mu. \quad (1.5)$$

The **variance** of x is

$$\text{VAR}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2, \quad (1.6)$$

$$= \int_{-\infty}^{\infty} p(x) \cdot x^2 dx - \mu^2, \quad (1.7)$$

$$= \int_{-\infty}^{\infty} \frac{x^2}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x - \mu)^2}{2\sigma^2}\right] dx - \mu^2, \quad (1.8)$$

$$= \sigma^2. \quad (1.9)$$

Remark 1.1.1. For the normal distribution, the mean μ is also equal to the median and mode. The skew and kurtosis are both zero.

Remark 1.1.2. The normal distribution is also called the **Gaussian distribution**. See Stahl¹ for an account for how *normal* arose, and how this naming “... had the disadvantage of leading people to believe that all other distributions of frequency are in one sense or another *abnormal*.”

1.2 Expectation

- Given a series of outcomes, expectation is a **measure** of that series, which will be shaped by the probability of each of the outcomes. Expectation is also called the weighted average.
- In the special case of equally likely probability for each outcome, the expectation will be the mean value, also called the simple average.

1.2.1 Non-Equal Likelihood

Let X be a random variable with a finite number of **outcomes** x_i for $i = 1, \dots, n$ with probability p_i . The **expectation** of X is defined as

$$\mathbb{E}[X] \triangleq \sum_{i=1}^n p_i x_i. \quad (1.10)$$

Because the sum of all probabilities is identically equal to unity,

$$\sum_{i=1}^n p_i = 1.0, \quad (1.11)$$

¹Stahl S. *The Evolution of the Normal Distribution*, Mathematics Magazine, 79:2, April 2006.

and the expected value is considered the **weighted average**, where each p_i is the **weight**.

Remark 1.2.3. The weighted average is also known as the **first moment**, which is analogous to the **center of mass** of a rigid body.

1.2.2 Equal Likelihood

For the special case where each probability is equally likely,

$$p_1 = p_2 = \cdots = p_n = \frac{1}{n}, \quad (1.12)$$

the expectation of $\mathbb{E}[X]$ has the specialized form μ ,

$$\mu \triangleq \mathbb{E}[X] \stackrel{(1.12)}{=} \frac{1}{n} \sum_{i=1}^n x_i. \quad (1.13)$$

and is referred to as the **simple average**.

Remark 1.2.4. The expectation $\mathbb{E}[X]$, also known as the weighted average of X , should not be confused with μ , the simple average of X . The latter is a special case of the former, as just demonstrated in the above sections.

Example 1. Consider a 6-sided dice, with faces 1 to 6. Consider the dice to be a *fair* dice, so that the probability of rolling a 1 is the same as rolling a 2, and is the same as rolling any of the other remaining numbers 3 to 6. Thus, the fair dice has equal likelihood of any of the 6 outcomes, and Eq. (1.13) is used. The expected value is calculated as

$$\mu = \mathbb{E}[X] = \frac{1}{6} (1 + 2 + 3 + 4 + 5 + 6) = 3.5 \quad (1.14)$$

The 6-sided dice roll in the preceding example can be illustrated through numerical simulation, implemented in Python below. The results from the Python script are shown in Fig. 1.1.

```

1  import matplotlib.pyplot as plt
2  from matplotlib import rc
3  rc('font',**{'family':'serif','serif':['Computer Modern Roman']})
4  rc('text', usetex=True)
5  import numpy as np
6  from random import randint  # random integer
7
8  n_rolls = 40  # number of dice rolls
9  roll_number = np.arange(1, n_rolls + 1)
10 roll_history = np.zeros(n_rolls)  # each roll result recorded in time
11 average = np.zeros(n_rolls)  # initialize the average, updated after each roll
12
13 for roll in roll_number:
14     roll_result = randint(1,6)  # roll a fair dice, values [1, 6]
15     print(f'Dice roll number {roll} roll value is {roll_result}.')
16     roll_history[roll-1] = int(roll_result)
17     sum_after_roll = sum(i for i in roll_history)
18     average[roll-1] = sum_after_roll / roll
19
20 fig = plt.figure(figsize=(8, 4))  # 8 inches wide, 4 inches tall
21
22 ax = fig.add_subplot(1, 1, 1)
23 ax.plot(roll_number, roll_history, 's', color='blue', label='roll value')
24 ax.plot(roll_number, average, color='red', marker='o', alpha=0.5,
25         label='roll average')
26 ax.grid()
27 ax.set_xlabel('roll number')
28 ax.set_ylabel('roll outcome and cumulative simple average')
29 ax.set_ylim(0, 7)
30 ax.legend(loc='upper right')
31 plt.show()
32 figure_name = 'dice_roll'
33 fig.savefig(figure_name + '.pdf', bbox_inches='tight')

```

The first moment analog to the center of mass can be illustrated through graphical inspection. Fig. 1.2 shows the roll outcomes, $1, \dots, 6$, along the x -axis and the probability of any of the six outcomes, $p_i = 1/6$, normalized to unity (since all of the six probabilities are equal) on the y -axis.

The center of mass of the six blue squares is readily seen at the $x = 3.5$ coordinate, indicated with a red circle. The center of mass is the simple average. This

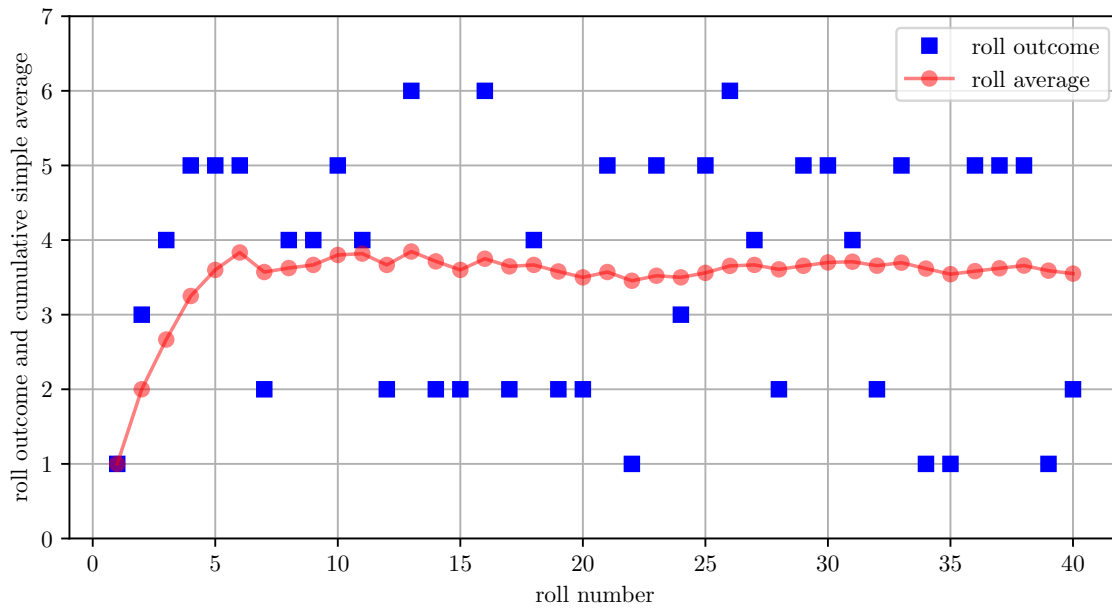


Figure 1.1: Numerical simulation of 40 dice rolls, showing value of each roll and average for all rolls. As the number of rolls increases, the average converges to 3.5, the dice roll expected value.

illustrates the equal probability special case, interpreted in the mechanical analog as the uniform density case.

For the general case with non-equal probability, *e.g.*, an unfair or weighted dice, the higher (or lower) probability of a particular outcome is analogous to an increase (or decrease) of local density near the particular x -coordinate of that outcome, and hence as the non-uniform density case.

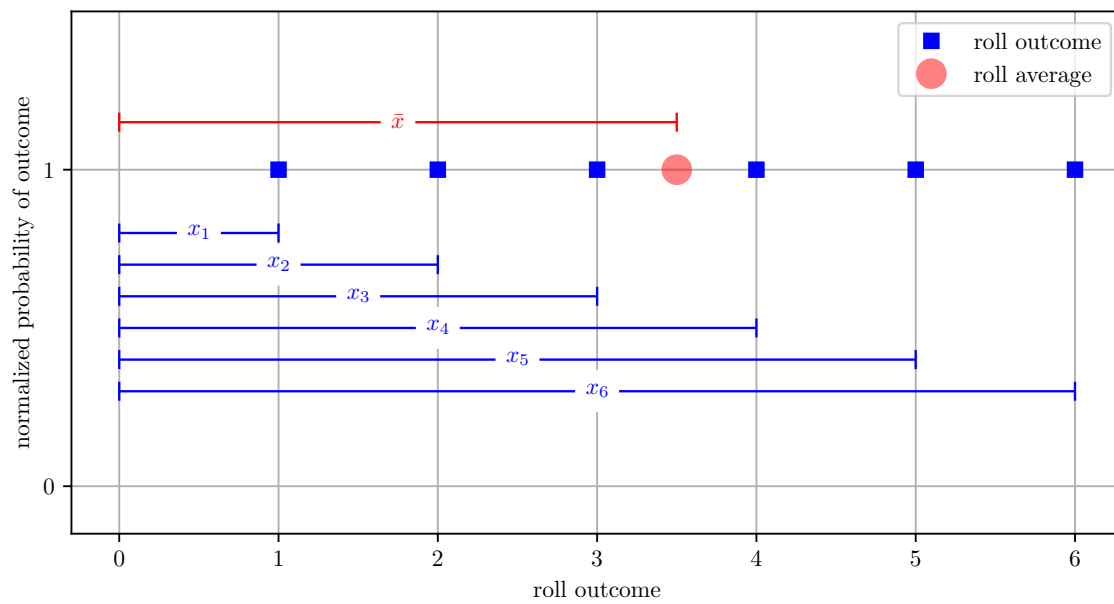


Figure 1.2: Graphical interpretation of average as first moment center of mass.

1.3 Variance

1.3.1 Non-Equal Likelihood

The **Variance** of X is defined as

$$\text{VAR}(X) \triangleq \mathbb{E}[(X - \mathbb{E}[X])^2], \quad (1.15)$$

$$\begin{aligned} & \mathbb{E}[(X - \mathbb{E}[X])(X - \mathbb{E}[X])], \\ & \mathbb{E}[X^2 - 2\mathbb{E}[X]X + \mathbb{E}[X]\mathbb{E}[X]], \\ & \mathbb{E}[X^2] - 2\mathbb{E}[X]\mathbb{E}[X] + \mathbb{E}[X]\mathbb{E}[X], \\ & \mathbb{E}[X^2] - (\mathbb{E}[X])^2. \end{aligned} \quad (1.16)$$

1.3.2 Equal Likelihood

For the special case where each probability is equally likely, (1.16) takes the form

$$\sigma^2 \triangleq \mathbb{E}[X] \stackrel{(1.16)}{=} \sum_{i=1}^n p_i x_i^2 - \mu^2 = \frac{1}{n} \sum_{i=1}^n x_i^2 - \mu^2, \quad (1.17)$$

and is referred to as the **simple variance**.

Remark 1.3.5. The form of (1.17) is often stated in slightly different form,

$$\frac{1}{n} \sum_{i=1}^n x_i^2 - \mu^2 = \frac{1}{n} \sum_{i=1}^n x_i^2 - 2\mu^2 + \mu^2, \quad (1.18)$$

$$\stackrel{(1.13)}{=} \frac{1}{n} \sum_{i=1}^n x_i^2 - \frac{2\mu}{n} \sum_{i=1}^n x_i + \frac{n\mu^2}{n}, \quad (1.19)$$

$$= \frac{1}{n} \sum_{i=1}^n (x_i^2 - 2\mu x_i + \mu^2), \quad (1.20)$$

$$= \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2. \quad (1.21)$$

1.4 Covariance

1.4.1 Non-Equal Likelihood

For two jointly distributed real-valued random variables X and Y , with **expected values**² $\mathbb{E}[X]$ and $\mathbb{E}[Y]$, respectively, the **covariance** is defined as

$$\text{Cov}(X, Y) \triangleq \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])], \quad (1.22)$$

$$\mathbb{E}[XY] - \mathbb{E}[X] \mathbb{E}[Y] - \mathbb{E}[X] \mathbb{E}[Y] + \mathbb{E}[X] \mathbb{E}[Y], \quad (1.23)$$

$$\mathbb{E}[XY] - \mathbb{E}[X] \mathbb{E}[Y]. \quad (1.24)$$

Let the random variable pair (X, Y) take on values (x_i, y_i) for $i = 1, \dots, n$ with probability p_i . The covariance is defined as

$$\text{Cov}(X, Y) \triangleq \sum_{i=1}^n p_i (x_i - \mathbb{E}[X]) (y_i - \mathbb{E}[Y]). \quad (1.25)$$

1.4.2 Non-Equal Likelihood

If the probability of each outcome is equal, the $p_i = 1/n$, and (1.25) becomes

$$\text{Cov}(X, Y) = \frac{1}{n} \sum_{i=1}^n (x_i - \mathbb{E}[X]) (y_i - \mathbb{E}[Y]). \quad (1.26)$$

1.4.3 Pearson Correlation Coefficient

The **Pearson correlation coefficient** (PCC)³ is a measure of the linear correlation between two variables X and Y .

When a pair of random variables (X, Y) are applied to a population, the PCC is called the **population correlation coefficient** and denoted as $\rho_{X,Y}$ and is defined as

$$\rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}, \quad (1.27)$$

where

- COV is the covariance of X and Y ,

²Also known as **mean values**.

³Also known as the Pearson's r coefficient, Pearson product-moment correlation coefficient (PPMCC), and the bivariate correlation.

- σ_X is the standard deviation of X , and
- σ_Y is the standard deviation of Y .

The PCC is bounded as

$$-1 \leq \rho_{X,Y} \leq 1. \quad (1.28)$$

- A value of -1 indicates complete negative linear correlation.
- A value of 0 indicates the absence of linear correlation.
- A value of +1 indicates complete positive linear correlation.

Remark 1.4.6. The covariance of a variable with itself is simply the variance of the variable. So, $\text{Cov}(X, X) = \text{Var}(X)$.

Remark 1.4.7. When $\mathbb{E}[XY] \approx \mathbb{E}[X] \mathbb{E}[Y]$, then the $\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X] \mathbb{E}[Y]$ is prone to so-called “catastrophic cancellation” with floating point calculations. A numerically stable algorithm should be used.

1.5 Cross-Correlation (Series Dot Product)

The cross-correlation is a measure of the *similarity in time* of two time series signals. It is often referred to as the *dot product* of two time series because the cross-correlation operation is a dot production operation.

1.5.1 Vector Dot Product

Consider two vectors $\mathbf{a}$ and $\mathbf{b}$ in three-dimensional space, measured from the same origin and with the same coordinate system, with components,

$$\mathbf{a} = [a_1, a_2, a_3], \quad (1.29)$$

$$\mathbf{b} = [b_1, b_2, b_3]. \quad (1.30)$$

$$(1.31)$$

The dot product of the two vectors is simply,

$$\mathbf{a} \cdot \mathbf{b} = a_i b_i = a_1 b_1 + a_2 b_2 + a_3 b_3 \quad (1.32)$$

$$\text{or generally } = \sum_{i=1}^n a_i b_i. \quad (1.33)$$

Next, consider $\mathbf{a}(t)$ and $\mathbf{b}(t)$ be two time-varying signals, occurring over time interval $[t_0, t_f]$. Let the amplitudes of $\mathbf{a}(t)$ and $\mathbf{b}(t)$ be measured from zero to t_f at n equally distributed points such that the signals are sampled at n uniform discrete times in the interval. The two vectors are written explicitly as

$$\mathbf{a} = [a_0, a_1, a_2, \dots, a_{n-3}, a_{n-2}, a_{n-1}], \quad (1.34)$$

$$\mathbf{b} = [b_0, b_1, b_2, \dots, b_{n-3}, b_{n-2}, b_{n-1}]. \quad (1.35)$$

Remark 1.5.8. To account for the *initial condition* of the time series, we decrement the counter,

$$1 \dots n, \quad (1.36)$$

to become

$$0 \dots n - 1. \quad (1.37)$$

In both of these cases, the number of sample points is identical, but the indexing is different by one.

Remark 1.5.9. Notice that the two signals $\mathbf{a}(t)$ and $\mathbf{b}(t)$ are *synchronized* in time such that

$$\mathbf{t} = [t_0, t_1, t_2, \dots, t_{n-3}, t_{n-2}, t_{n-1}]. \quad (1.38)$$

Notice that $\Delta t = t_1 - t_0 = t_2 - t_1 = \dots t_{n-1} - t_{n-2}$ and that the signals are sampled at a frequency of $\frac{1}{\Delta t}$ Hz.

1.5.2 Series Dot Product

The dot product of these two signals is

$$\mathbf{a} \cdot \mathbf{b} = a_i b_i = a_0 b_0 + a_1 b_1 + a_2 b_2 + \dots + a_{n-1} b_{n-1} \quad (1.39)$$

$$\text{or generally} = \sum_{i=0}^{n-1} a_i b_i. \quad (1.40)$$

For comparison of two time series, the dot product will be referred to as the **cross-correlation score**.

Example 2. This example is a reproduction of the cross-correlation example from <https://anomaly.io/understand-auto-cross-correlation-normalized-shift/>. Let vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ be defined as follows,

$$\mathbf{a} = [1, 2, -2, 4, 2, 3, 1, 0] \quad (1.41)$$

$$\mathbf{b} = [2, 3, -2, 3, 2, 4, 1, -1] \quad (1.42)$$

$$\mathbf{c} = [-2, 0, 4, 0, 1, 1, 0, -2] \quad (1.43)$$

Then the cross-correlation score between $\mathbf{a}$ and $\mathbf{b}$ is found to be high,

$$\mathbf{a} \cdot \mathbf{b} = 41; \quad (1.44)$$

whereas, the cross-correlation score between $\mathbf{a}$ and $\mathbf{c}$ is found to be low,

$$\mathbf{a} \cdot \mathbf{c} = -5. \quad (1.45)$$

1.5.3 Limitation of Cross-Correlation

A significant limitation of the cross-correlation method is its dependence on *amplitude*:

- Large positive magnitudes increase the cross-correlation.
- Large negative magnitudes decrease the cross-correlation.
- A cross-correlation score shouldn't get better (positive) or worse (negative) if the amplitude of the signal is increased or decreased.
- Shape similarity is obfuscated by amplitude dependence. For example, the three cross-correlation scores generated from comparison of three signals, $A \sin(t)$, $B \sin(t)$, and $C \sin(t)$, all of the same shape, $\sin(t)$, but with unique amplitudes A, B, C , will report that two out of the three signals are better correlated by virtue of their closer amplitudes. In a sense, this result is satisfactory, since amplitude distinguishes one signal from another. However, this result is incomplete, since the amplitude dependence obfuscates the fact that all three signals have the same basic shape.
- Shape similarity obfuscation is easily seen with shape *self*-similarity, that is, using the *same* signal for both vectors in the cross-correlation. In this case

$$\mathbf{a} \cdot \mathbf{a} > \mathbf{a} \cdot \frac{\mathbf{a}}{2}, \quad (1.46)$$

which is a sensible outcome, but not helpful to distinguish shape similarity.

To remedy the shortcoming of cross-correlation caused by amplitude dependence, **vector normalization** is employed, as explained in a following section. Prior to this, however, we make a brief introduction to the p-norm.

1.5.4 p-norm

For any any number $p \in \mathbb{R}$ and $p \geq 1$, the **p-norm** is defined as

$$\|\mathbf{a}\|_p \triangleq \left(\sum_{i=0}^{n-1} |a_i|^p \right)^{\frac{1}{p}}. \quad (1.47)$$

Example 3. The L2-norm, also called the **Euclidean norm**, is defined as

$$\| \mathbf{a} \|_2 \triangleq \left(\sum_{i=0}^{n-1} |a_i|^2 \right)^{\frac{1}{2}}, \quad (1.48)$$

and has the physical interpretation as the length of the vector on which the L2-norm operates.

The L1-norm, also called the **Manhattan taxicab norm**, is defined as

$$\| \mathbf{a} \|_1 \triangleq \left(\sum_{i=0}^{n-1} |a_i| \right), \quad (1.49)$$

and has the physical interpretation as the cumulative length (mileage) accumulated for each segment a_i .

The **absolute value** is the L1-norm on a one-dimensional vector space,

$$\| \mathbf{a} \|_1 \triangleq \left(\sum_{i=0}^0 |a_i| \right) = |a_0|, \quad (1.50)$$

1.5.5 Normalized Cross-Correlation

When vector normalization is used with cross-correlation, the resulting method is referred to as *normalized cross-correlation*. Normalized cross-correlation is the same as cross-correlation with one exception: The vectors used in the dot product are normalized by their respective L2-norm.

The normalized cross-correlation for the three-dimensional vectors $\mathbf{a}$ and $\mathbf{b}$ is defined as

$$\frac{\mathbf{a}}{\| \mathbf{a} \|_2} \cdot \frac{\mathbf{b}}{\| \mathbf{b} \|_2} \triangleq \frac{a_i}{\sqrt{\sum_{i=1}^n a_i a_i}} \cdot \frac{b_i}{\sqrt{\sum_{i=1}^n b_i b_i}} \quad (1.51)$$

$$= \frac{a_1 b_1 + a_2 b_2 + a_3 b_3}{\left(\sqrt{a_1^2 + a_2^2 + a_3^2} \right) \left(\sqrt{b_1^2 + b_2^2 + b_3^2} \right)}. \quad (1.52)$$

The foregoing result generalizes to time series $\mathbf{a}(t)$ and $\mathbf{b}(t)$ as

$$\frac{\mathbf{a}}{\|\mathbf{a}\|} \cdot \frac{\mathbf{b}}{\|\mathbf{b}\|} \triangleq \frac{\sum_{i=0}^{n-1} a_i b_i}{\left(\sqrt{\sum_{i=0}^{n-1} a_i^2} \right) \left(\sqrt{\sum_{i=0}^{n-1} b_i^2} \right)} \quad (1.53)$$

Consider again the vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ from Example 1.

Example 4. Consider again the vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ from Example 1. The *normalized* cross-correlation score between $\mathbf{a}$ and $\mathbf{b}$ is found to be high (relative to a maximum of 1.0),

$$\frac{\mathbf{a}}{\|\mathbf{a}\|} \cdot \frac{\mathbf{b}}{\|\mathbf{b}\|} = \frac{41}{\sqrt{39} \sqrt{48}} = 0.947; \quad (1.54)$$

whereas, the *normalized* cross-correlation score between $\mathbf{a}$ and $\mathbf{c}$ is found to be low (relative to no correlation for 0.0),

$$\frac{\mathbf{a}}{\|\mathbf{a}\|} \cdot \frac{\mathbf{c}}{\|\mathbf{c}\|} = \frac{-5}{\sqrt{39} \sqrt{26}} = -0.157. \quad (1.55)$$

The **benefits** of *normalized* cross-correlation are two-fold: Amplitude independence and unit normal scoring. Amplitude independence is seen as

$$\mathbf{a} \cdot \mathbf{a} = \mathbf{a} \cdot \frac{\mathbf{a}}{2}. \quad (1.56)$$

The normalized cross-correlation score now ranges between -1 and +1.

- +1 is perfect correlation,
- 0 is no correlation, and
- -1 is anti-correlation (correlated, but in opposite phase).

1.5.6 Normalized Cross-Correlation with Time Shift

The motivation for **normalization** was to remove amplitude effects from the cross-correlation score. These effects are seen along the y -axis. A complementary effect, called **phase effect**, can exist along the x -axis.

Specifically, two time series may be highly correlated, or even identical, but if out of phase, will cause the cross-correlation and the normalized cross-correlation to report a score of zero.

To remedy this outcome, a second strategy can be formulated. In previous sections, the dot product and the normalized dot product were used for cross-correlation and normalized cross-correlation, respectively. Now, we will shift one of the two signals incrementally in time to both the left and the right, to create what is referred to as the **sliding dot product** and **sliding normalized dot product**, respectively.

Conceptually, the sliding dot product strategy computes the dot product between two time series for all combination of a_i and b_i to determine if a **phase lead** or **phase lag** results in a better correlation score.

The result of the sliding dot product is an expansion of inner products, called **cross-correlation coefficients**, which will be labeled as vector $\mathbf{c}$, with dimension $2n - 1$, generated by the sliding mechanism. The sliding dot product is illustrated in matrix form, where time series $\mathbf{b}$ slides forward and backward incrementally in time, relative to time series $\mathbf{a}$. This operation takes the explicit form,

$$\begin{array}{c}
 \text{initial condition, len} = 1 \\
 \hline
 \text{slide-forward} \\
 \text{operation} \\
 \vdots \\
 \text{len} = n - 1 \\
 \hline
 \text{slide-backward} \\
 \text{operation} \\
 \vdots \\
 \text{len} = n - 1
 \end{array}
 \left\{ \begin{array}{c}
 c_0 \\
 c_1 \\
 c_2 \\
 \vdots \\
 c_{n-1} \\
 \hline
 c_{-1} \\
 c_{-2} \\
 \vdots \\
 c_{-(n-1)}
 \end{array} \right\} =$$

$(2n-1 \times 1)$

$$\left[\begin{array}{cccccc|cccc}
 b_0 & b_1 & b_2 & \cdots & b_{n-3} & b_{n-2} & b_{n-1} & & & \\
 0 & b_0 & b_1 & b_2 & \cdots & b_{n-3} & b_{n-2} & & & \\
 0 & 0 & b_0 & b_1 & b_2 & \cdots & b_{n-3} & & & \\
 & & & \ddots & & & & & & \\
 & & & & \ddots & & & & & \\
 & & & & & \ddots & & & & \\
 & & & & & & \ddots & & & \\
 0 & 0 & 0 & 0 & 0 & \cdots & b_0 & & & \\
 \hline
 b_1 & b_2 & \cdots & b_{n-3} & b_{n-2} & b_{n-1} & 0 & & & \\
 b_2 & \cdots & b_{n-3} & b_{n-2} & b_{n-1} & 0 & 0 & & & \\
 \vdots & & & \ddots & & & & & & \\
 \vdots & & & \ddots & & & & & & \\
 \vdots & \ddots & & & & & & & & \\
 b_{n-1} & 0 & 0 & 0 & 0 & 0 & 0 & & &
 \end{array} \right]_{(2n-1 \times n)} \quad \left\{ \begin{array}{c} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_{n-3} \\ a_{n-2} \\ a_{n-1} \end{array} \right\}_{(n \times 1)} \quad (1.57)$$

Example 5. This example is a reproduction of the normalized cross-correlation with time shift example from <https://anomaly.io/understand-auto-cross-correlation-normalized-shift/>. Let vectors $\mathbf{a}$ and $\mathbf{b}$ be defined as follows,

$$\mathbf{a} = [0, 1, 2, 3, 4, \quad 0, 1, 2, 3, 4, \quad 0, 1, 2, 3, 4, \quad 0, 1, 2, 3, 4] \quad (1.58)$$

$$\mathbf{b} = [1, 2, 3, 3, 0, \quad 1, 2, 3, 4, 0, \quad 1, 1, 4, 4, 0, \quad 1, 2, 3, 4, 0] \quad (1.59)$$

where slight spacing has been interjected between every fifth element, to help illustrate the pattern embedded in the series. Since the dimension, n , of the signal is 20, there are $2n - 1 = 39$ correlation coefficients, c_i , that can be calculated from the normalized cross-correlation with time shift operation. Below are 9 of the 39 correlation coefficients, showing the initial condition, the first four slide-forward operations, and the first four slide-backward operations.

Δt	-4	-3	-2	-1	0	1	2	3	4
c_i	0.862	0.021	-0.547	-0.423	0.000	0.867	0.127	-0.466	-0.393

The results indicate that sliding the $\mathbf{b}$ signal forward by one time increment results in the largest normalized cross-correlation with time shift correlation coefficient ($c_1 = 0.867$ at $\Delta t = 1$). This correlation can be seen through visual inspection of the two curves at zero time shift, and then shifting signal $\mathbf{b}$ forward by one step.

1.5.7 Continuous Case - Disregard this Section for Now

Finally, let the cross-correlation function $y(t)$ be the sliding inner product of the signal $x(t)$ with the weighting function $w(t)$ such that

$$y(t) = \int_0^{t_f} x(t) w(t - \tau) d\tau, \quad (1.60)$$

which is the filter **slide-forward option**, and

$$y(t) = \int_{-t_f}^0 x(t) w(t - \tau) d\tau \quad (1.61)$$

is the **slide-backward operation**. The two forgoing operations can be combined to generally be the **sliding dot product operation** as

$$y(t) = \int_{-t_f}^{t_f} x(t) w(t - \tau) d\tau. \quad (1.62)$$

1.5.8 From Deterministic to Stochastic

In this section, we apply some of the notions of expectation, variance, and covariance that we noted in the first sections with the notions of cross-correlation. Up until this point, the signals used in the cross-correlation discussion were tactically normalized to the origin at $\mathbf{0}$. So, expectations and deviations were tactically taken with respect to zero for all times in signals a_i and b_i .

Now, we note the form for correlation applied to stochastic processes (the former will more or less be called a deterministic process, by contrast). In the stochastic process, we will normalize data with respect to an expected value (e.g., mean, in the case of equal likelihood probabilities) as an inhomogeneous datum (different from zero). The mean *could* be zero as a trivial special case, but in general the mean is non-zero.

Stochastic Vector Dot Product

Generally, we substitute $(a_i - \bar{a})$ for (a_i) in the previous definition of vector dot product, appearing in Section 1.5.1. The value $\bar{a}$ is the expected value of $\mathbf{a}$, and in the special case of equal likelihood, $\bar{a}$ is the average value μ_a .

Then, the stochastic vector dot product of three-dimensional signals $\mathbf{a}$ and $\mathbf{b}$ is defined as,

$$\mathbf{a} \cdot \mathbf{b} = (a_i - \bar{a}) (b_i - \bar{b}) \quad (1.63)$$

$$= (a_1 - \bar{a}) (b_1 - \bar{b}) + (a_2 - \bar{a}) (b_2 - \bar{b}) + (a_3 - \bar{a}) (b_3 - \bar{b}) \quad (1.64)$$

$$\text{or generally} = \sum_{i=1}^n (a_i - \bar{a}) (b_i - \bar{b}). \quad (1.65)$$

Stochastic Series Dot Product

Extending this to n -dimensional signals $\mathbf{a}(t)$ and $\mathbf{b}(t)$, the stochastic series dot product becomes,

$$\mathbf{a} \cdot \mathbf{b} = (a_i - \bar{a}) (b_i - \bar{b}) \quad (1.66)$$

$$= (a_0 - \bar{a}) (b_0 - \bar{b}) + (a_1 - \bar{a}) (b_1 - \bar{b}) + (a_2 - \bar{a}) (b_2 - \bar{b}) + \dots + (a_{n-1} - \bar{a}) (b_{n-1} - \bar{b}) \quad (1.67)$$

$$\text{or generally} = \sum_{i=0}^{n-1} (a_i - \bar{a}) (b_i - \bar{b}). \quad (1.68)$$

Stochastic Series Normalized Dot Product

The normalization of the stochastic series dot product arises by comparison of the expected value to its variance. Noting, with the special case of equal likelihood,

$$\sigma^2 = \frac{1}{n} \sum_{i=0}^{n-1} (a_i - \bar{a})^2, \quad (1.69)$$

then,

$$\sigma = \sqrt{\frac{1}{n} \sum_{i=0}^{n-1} (a_i - \bar{a})^2}. \quad (1.70)$$

This allows the stochastic series normalized dot product to be written as

$$\frac{\mathbf{a}}{\|\mathbf{a}\|_2} \cdot \frac{\mathbf{b}}{\|\mathbf{b}\|_2} \triangleq \frac{\sum_{i=0}^{n-1} (a_i - \bar{a}) (b_i - \bar{b})}{(n) \left(\sqrt{\sum_{i=0}^{n-1} (a_i - \bar{a})^2} \right) \left(\sqrt{\sum_{i=0}^{n-1} (b_i - \bar{b})^2} \right)}. \quad (1.71)$$

Some statistical references call this the **zero-normalized cross-correlation**.

1.6 Error and Error Rate

1.6.1 Error

The **error** $\mathbf{e}(t)$ between two signals $\mathbf{a}(t)$ and $\mathbf{b}(t)$ is defined as the difference between the two signals,

$$\mathbf{e}(t) \triangleq |\mathbf{a}(t) - \mathbf{b}(t)|. \quad (1.72)$$

Note that $\mathbf{e} \geq \mathbf{0} \ \forall \ t$. If $\mathbf{b}$ exactly follows $\mathbf{a}$ at every time step, then the error is zero, as expected.

Example 6. Let vectors $\mathbf{a}$ and $\mathbf{b}$ be defined as follows,

$$\mathbf{a} = [20, 20, 20, 20] \quad (1.73)$$

$$\mathbf{b} = [18, 18, 18, 18] \quad (1.74)$$

Then the error $\mathbf{e}$ is found to be,

$$\mathbf{e} = [2, 2, 2, 2] \quad (1.75)$$

In this example, assuming $\mathbf{a}$ is the ground truth, $\mathbf{b}$ underestimates $\mathbf{a}$ by 2 at every sample. This is an example of **systemic error**, where there is a consistent offset between the two signals, and an offset adjustment (the offset value of 2 in this case) would eliminate all error between the two signals.

1.6.2 Accumulated Error

Next, we would like to have a scalar value to describe the error over the entire sample, which is measured at each sample point and has dimension n .

We define the **L1-norm accumulated error**, $\|\mathbf{e}\|_1$, as the sum of the errors of each of the measurement,

$$\|\mathbf{e}\|_1 \triangleq \sum_{i=0}^{n-1} |e_i| = \sum_{i=0}^{n-1} |a_i - b_i|. \quad (1.76)$$

This is the L1-norm, applied to each error measurement, $e_i = a_i - b_i$. A serious problem with accumulated error is that the error increases the more times the signals are measured.

Example 7.

- In the preceding example, if the signals were measured at only one sample, the accumulated error would be 2.
- Similarly, if two samples were taken, the accumulated error would be 4.
- For three samples, the accumulated error would be 6.
- For the example as stated above, with four samples, the accumulated error is 8.

This pattern shows the limitation of accumulated error: that it is dependent on sample size.

We define the **L2-norm accumulated error**, $\| \mathbf{e} \|_2$, as the sum of the errors of each of the measurement,

$$\| \mathbf{e} \|_2 \triangleq \left(\sum_{i=0}^{n-1} |e_i|^2 \right)^{\frac{1}{2}} = \left(\sum_{i=0}^{n-1} |a_i - b_i|^2 \right)^{\frac{1}{2}}. \quad (1.77)$$

This is the L2-norm, applied to the error measurement, $e_i = a_i - b_i$. The systemic error seen above for the L1-norm, that the error increases with increasing number of measurements, also occurs for the L2-norm. To remedy this effect, normalization by the sample count, called **error rate**, will be used.

1.6.3 Error Rate

The concept of **error rate** is to normalize the accumulated error for any Lp -norm type by the dimension $n = \text{len}(\mathbf{a}) = \text{len}(\mathbf{b})$ of the signals. The resulting metric then reports error on a *per-measurement basis*, which eliminates the accumulation of error. Specifically, the **L1-norm error rate** is defined as

$$\| \bar{\mathbf{e}} \|_1 \triangleq \frac{1}{n} \sum_{i=0}^{n-1} |e_i| = \frac{1}{n} \sum_{i=0}^{n-1} |a_i - b_i|, \quad (1.78)$$

and the **L2-norm error rate** is defined as

$$\| \bar{\mathbf{e}} \|_2 \triangleq \frac{1}{n} \left(\sum_{i=0}^{n-1} |e_i|^2 \right)^{\frac{1}{2}} = \frac{1}{n} \left(\sum_{i=0}^{n-1} |a_i - b_i|^2 \right)^{\frac{1}{2}}. \quad (1.79)$$

Notice that the error rate units are [amplitude/sample], hence the rate of the errors in the amplitudes.

Example 8. Revising the example with vectors $\mathbf{a}$ and $\mathbf{b}$,

$$\mathbf{a} = [20, 20, 20, 20] \quad (1.80)$$

$$\mathbf{b} = [18, 18, 18, 18] \quad (1.81)$$

The **L1-norm error rate** $\|\bar{\mathbf{e}}\|_1$ is found to be,

$$\|\bar{\mathbf{e}}\|_1 = \frac{1}{4} \cdot (2 + 2 + 2 + 2) = 2. \quad (1.82)$$

The **L2-norm error rate** $\|\bar{\mathbf{e}}\|_2$ is found to be,

$$\|\bar{\mathbf{e}}\|_2 = \frac{1}{4} \cdot \sqrt{(4 + 4 + 4 + 4)} = 1. \quad (1.83)$$